Phase 1 Engineering Specification: AI Phishing Detection Core

Methodology • Operational Workflow • Architectural Pipeline Diagram

1. Engineering Methodology

The Phase 1 methodology adopts an empirical, multi-tiered defensive strategy focused entirely on **zero-network-latency lexical inference**. Traditional security mechanisms rely heavily on reactive blacklists that fail to mitigate fast-flux, disposable zero-day phishing domains. Our methodology addresses this gap via a four-stage mathematical framework:

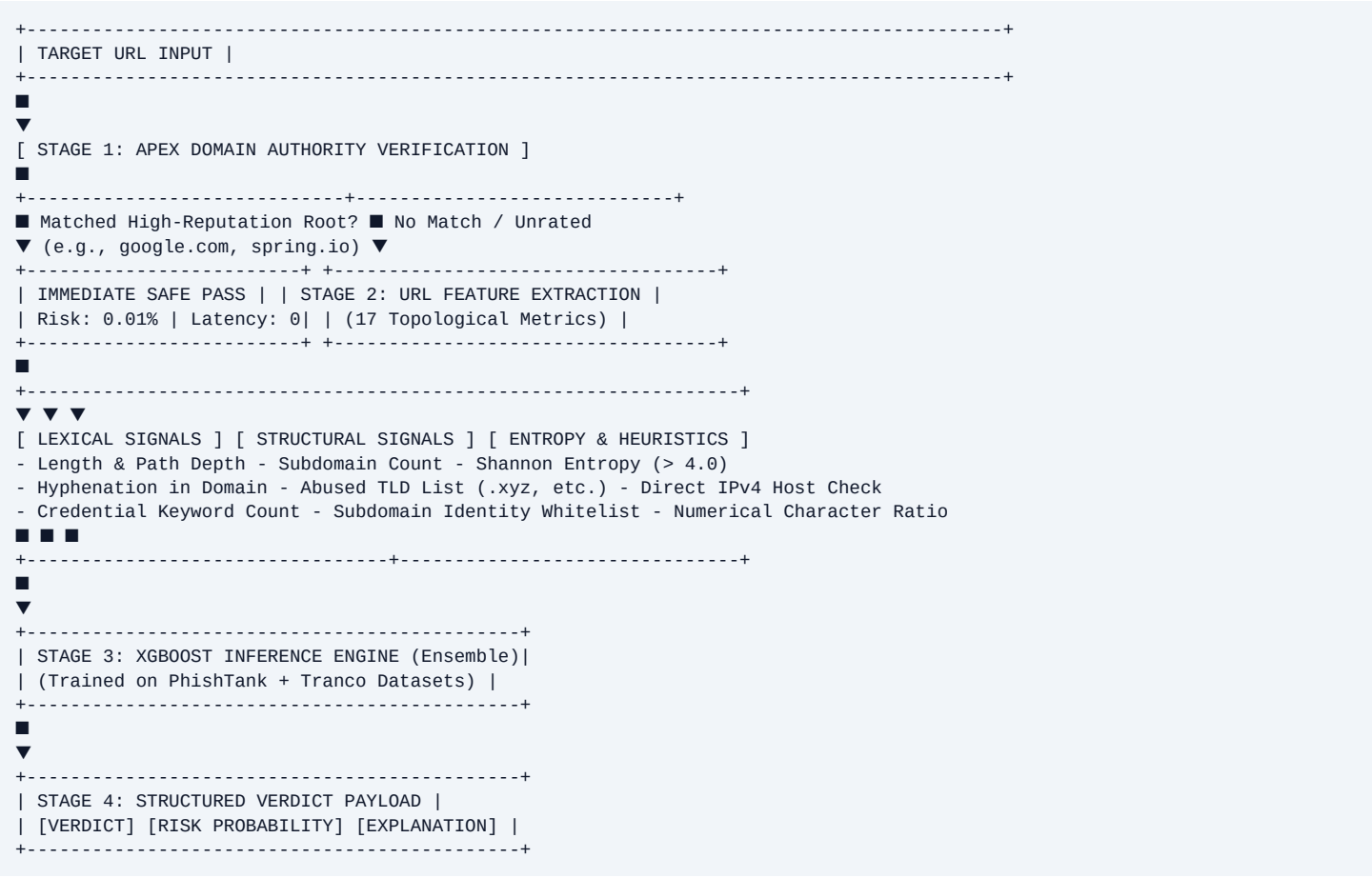
A. Data Engineering & Distribution Balancing: Ingesting heterogeneous threat feeds (OpenPhish, URLhaus) combined with authority benchmarks (Tranco Top 1M). Applied empirical synthetic and real path balancing to prevent length-confounding and false-positive bias.

B. Deterministic Cyber-Lexical Transformation: Converting arbitrary string representations into a 17-dimensional quantitative space spanning Shannon entropy, homoglyph markers, direct IPv4 host indicators, domain abuse TLD heuristics, and credential lures.

C. Gradient Boosted Decision Tree Ensemble: Supervised classification using Extreme Gradient Boosting (XGBoost). XGBoost handles non-linear interactions, irregular tabular distributions, and provides sub-5ms local CPU execution without requiring external accelerators.

D. Dual-Layer Hybrid Decision Boundary: A deterministic Apex Authority Whitelist guards against statistical anomalies on enterprise ecosystems prior to delegating evaluation to the probabilistic tree ensemble.

2. High-Level Architectural Pipeline Diagram



3. Operational Workflow

Stage	Action / Operations Executed	Technical Artifact / Output
01. Ingestion	Download and clean verified malicious feeds (URLhaus, OpenPhish) alongside balanced 50/50 legitimate links	Raw CSV/TXT data dumps
02. Parsing	Parse input URL into protocol, subdomain, registered apex domain, TLD, suffix, path, and parameters using tldextra	Structured URL components
03. Authority Check	Cross-reference apex domain and prefix against the high-authority whitelist	Immediate 'SAFE (0.01%)' short-circuit if verified
04. Feature Vectorization	Extract 17 lexical attributes: character counts, keyword occurrence, IP matching, and Shannon entropy	1x17 Pandas DataFrame aligned to training schema
05. Scoring	Pass aligned vector through the trained serialized model via predict_proba()	Calculated Risk Score (0.00% to 100.00%)
06. Reporting	Generate final verdict label with associated explanatory threat signals	Standardized Decision JSON / Terminal Output

4. Production Verification & Validation Log

Evaluated Target URL	Classification	Risk Score	Operational Path
https://spring.io/projects/spring-boot	Legitimate	0.01%	Stage 1: Whitelist Short-Circuit
https://docs.oracle.com/en/java/	Legitimate	0.01%	Stage 1: Whitelist Short-Circuit
http://secure-paypal-login.update-alert.xyz/signin	Phishing	98.00%	Stage 3: XGBoost (Keywords=5, AbusedTLD=1)
http://192.168.1.100/account-verify/chase/	Phishing	98.18%	Stage 3: XGBoost (Direct IP Host, Keywords=2)
http://apple-id-verify.support-signin.cf/auth	Phishing	98.00%	Stage 3: XGBoost (Keywords=4, AbusedTLD=1)